GATE ME 90 Days Challenge

Day 42/90 · Practice Set with Detailed Solutions

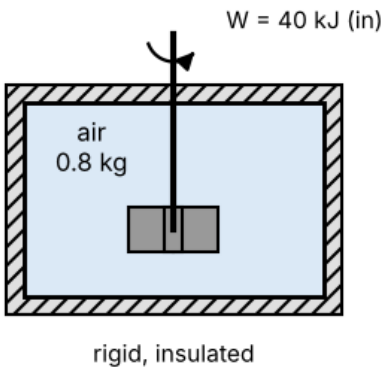
SUBJECT	TODAY'S TOPIC	COUNTDOWN
Weekly Revision	Week 7 Revision	63 days to GATE

How to use this set: 15 questions (7 MCQ, 8 NAT) on all five topics of Week 7: first law for closed systems and control volumes; second law and entropy; differential equations; availability and thermodynamic relations; 1-D steady conduction (3 questions each). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

A rigid, insulated tank holds 0.8 kg of air (c_v = 0.718 kJ/kg·K). A paddle wheel driven from outside, as shown, does 40 kJ of work on the air. The rise in air temperature (in K, round off to two decimal places) is _____



Q2 MCQ

A closed system goes from state 1 to state 2 along different paths. Which quantity is the same for every path?

(A) Q
(B) W
(C) Q − W
(D) Q + W

Q3 NAT

Air enters an adiabatic diffuser at 250 m/s and leaves at 30 m/s (c_p = 1.005 kJ/kg·K). The rise in air temperature (in K, round off to two decimal places) is _____

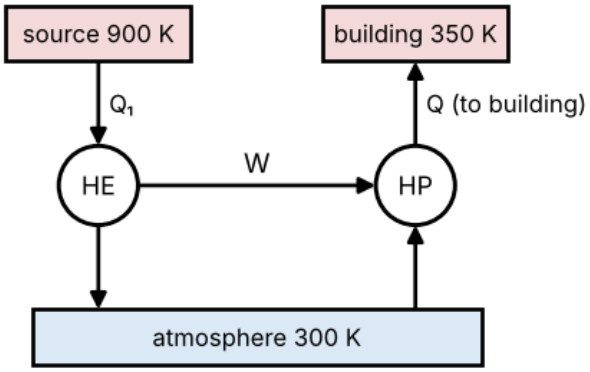
Q4 MCQ

For an adiabatic process in a closed system, the entropy of the system

(A) always remains constant
(B) may decrease if the process is irreversible
(C) remains constant if reversible and increases if irreversible
(D) always decreases

Q5 NAT

A reversible heat engine takes heat from a source at 900 K and rejects heat to the atmosphere at 300 K. All its work drives a reversible heat pump that takes heat from the atmosphere at 300 K and delivers it to a building at 350 K, as shown. The heat delivered to the building per kJ of heat taken from the 900 K source (round off to two decimal places) is _____



Q6 NAT

1 kg of ice at 0 °C melts completely to water at 0 °C (latent heat 333.5 kJ/kg). The entropy change of the ice (in kJ/K, round off to three decimal places) is _____

Q7 MCQ

A particular integral of $y'' - 4y' + 4y = e^{2x}$ is

(A) $e^{2x}/4$
(B) $x\, e^{2x}$
(C) $(x^2/2)\, e^{2x}$
(D) $x^2\, e^{2x}$

Q8 NAT

The solution of $dy/dx = -2xy$ with $y(0) = 3$ is evaluated at $x = 1$. The value of $y(1)$ (round off to three decimal places) is _____

Q9 MCQ

If F(s) is the Laplace transform of $f(t) = e^{-2t} \cos 3t$, the value of F(1) is

(A) 1/6
(B) 1/10
(C) 1/5
(D) 1/3

Q10 NAT

A room is kept at 22 °C by an electric resistance heater when the outdoor temperature is 2 °C. Treating the outdoors as the dead state, the second-law efficiency of the heater (in %, round off to two decimal places) is _____

Q11 MCQ

For an incompressible liquid, the difference between the specific heats $c_p - c_v$ is

(A) R
(B) zero
(C) T v β^2/κ, which is large
(D) equal to c_v

Q12 MCQ

In a process the total entropy generated is 0.4 kJ/K. The surroundings are at 27 °C. The lost work (irreversibility) is

(A) 10.8 kJ
(B) 120.1 kJ
(C) 0.4 kJ
(D) 750.4 kJ

Q13 NAT

A furnace wall has 0.2 m of firebrick (k = 1.2 W/m·K) on the inside and 0.1 m of insulating brick (k = 0.15 W/m·K) outside it. The inner face is at 900 °C and the outer face at 60 °C. The temperature at the firebrick–insulation interface (in °C) is _____

Q14 MCQ

A bare steam pipe has an outer radius larger than the critical radius of insulation. Adding insulation to it will

(A) first increase and then decrease the heat loss
(B) always decrease the heat loss
(C) always increase the heat loss
(D) not change the heat loss

Q15 NAT

A spherical liquid-nitrogen vessel has an inner radius of 0.5 m and is covered with 50 mm of insulation (k = 0.02 W/m·K). The inner surface is at −196 °C and the outer surface at 25 °C. The heat gain (in W, round off to one decimal place) is _____

Answer Key

Q	Type	Answer
Q1	NAT	69.64 (accept 69.0 to 70.3)
Q2	MCQ	(C) Q − W
Q3	NAT	30.65 (accept 30.3 to 31.0)
Q4	MCQ	(C) remains constant if reversible and increases if irreversible
Q5	NAT	4.67 (accept 4.62 to 4.71)
Q6	NAT	1.221 (accept 1.209 to 1.233)
Q7	MCQ	(C) $(x^2/2)\, e^{2x}$
Q8	NAT	1.104 (accept 1.093 to 1.115)
Q9	MCQ	(A) 1/6
Q10	NAT	6.78 (accept 6.71 to 6.85)
Q11	MCQ	(B) zero
Q12	MCQ	(B) 120.1 kJ
Q13	NAT	732
Q14	MCQ	(B) always decrease the heat loss
Q15	NAT	305.5 (accept 302.4 to 308.6)

Detailed Solutions

Q1 **Answer: 69.64 (accept 69.0 to 70.3)**

Concept: First law, closed system: $Q - W = \Delta U$. Rigid tank → no boundary work, but shaft (paddle) work is still work.

$Q = 0$, $W = -40$ kJ (work done on the air), so $\Delta U = +40$ kJ

$\Delta T = 40/(0.8 \times 0.718) =$ **69.64 K**

Exam tip: 'Rigid' kills only p dV work; paddle-wheel and electrical work still cross the boundary.

Q2 **Answer: (C) Q - W**

Concept: Q and W are path functions (inexact differentials), but their difference equals ΔU.

Since U is a property, $Q - W = U_2 - U_1$ is fixed by the end states alone.

Exam tip: This is the first law's corollary: energy (U) is a property of the system.

Q3 **Answer: 30.65 (accept 30.3 to 31.0)**

Concept: SFEE with $Q = W = 0$ and $\Delta z = 0$: $h_1 + V_1^2/2 = h_2 + V_2^2/2$.

$\Delta T = (V_1^2 - V_2^2)/(2c_p) = (62\,500 - 900)/(2 \times 1005)$

$\Delta T =$ **30.65 K**

Exam tip: Use c_p in J/kg·K (1005) when velocities are in m/s.

Q4 Answer: (C) remains constant if reversible and increases if irreversible

Concept: $\Delta S = \int \delta Q/T + S_{gen}$. With $\delta Q = 0$, $\Delta S = S_{gen} \geq 0$.

Reversible adiabatic (isentropic): $S_{gen} = 0$, $\Delta S = 0$. Irreversible adiabatic (friction, stirring): $\Delta S > 0$.

Entropy of an adiabatic closed system can never decrease.

Exam tip: 'Adiabatic' ≠ 'isentropic' unless the process is also reversible.

Q5 Answer: 4.67 (accept 4.62 to 4.71)

Concept: Chain the two devices through their common work: $W = \eta_{HE} Q_1$ and $Q_{building} = COP_{HP} \times W$.

$\eta_{HE} = 1 - 300/900 = 0.6667$; $COP_{HP} = 350/(350 - 300) = 7$

$Q_{building}/Q_1 = 0.6667 \times 7 = \mathbf{4.67}$

Exam tip: More heat reaches the building than leaves the source: the heat pump pulls extra energy from the atmosphere.

Q6 Answer: 1.221 (accept 1.209 to 1.233)

Concept: Melting at constant temperature is internally reversible: $\Delta S = Q/T$ with T in kelvin.

$\Delta S = 333.5/273.15$

$\Delta S = \mathbf{1.221\ kJ/K}$

Exam tip: Using 0 °C instead of 273.15 K is the classic mistake (it would divide by zero).

Q7 Answer: (C) $(x^2/2)\, e^{2x}$

Concept: The auxiliary equation $(m - 2)^2 = 0$ has a double root 2, which matches the exponent of the RHS; so try $y_p = A\, x^2\, e^{2x}$.

For $(D - 2)^2 y = e^{2x}$: $y_p = x^2 e^{2x}/2!$ (rule for a root repeated twice)

Check: with $y = (x^2/2)e^{2x}$, $(D - 2)y = x\, e^{2x}$ and $(D - 2)^2 y = e^{2x}$ ✓, so $y_p = \mathbf{(x^2/2)\, e^{2x}}$

Exam tip: If e^{ax} on the RHS repeats a root k times, multiply the trial solution by x^k.

Q8 Answer: 1.104 (accept 1.093 to 1.115)

Concept: Variables separable: $dy/y = -2x\, dx$.

$\ln y = -x^2 + C \Rightarrow y = 3e^{-x^2}$ using $y(0) = 3$

$y(1) = 3e^{-1} = \mathbf{1.104}$

Exam tip: Always apply the initial condition before substituting the final x.

Q9 Answer: (A) 1/6

Concept: First shifting theorem: $L\{e^{-at} g(t)\} = G(s + a)$, with $L\{\cos bt\} = s/(s^2 + b^2)$.

$F(s) = (s + 2)/[(s + 2)^2 + 9]$

$F(1) = 3/(9 + 9) = 3/18 = \mathbf{1/6}$

Exam tip: Shift s, not t: replacing s by $s - 2$ instead of $s + 2$ gives $(-1)/(1 + 9) = -1/10$, a common slip.

Q10 **Answer: 6.78 (accept 6.71 to 6.85)**

Concept: η_{II} = minimum work needed / actual work used. The minimum is a reversible heat pump between 275.15 K and 295.15 K.

COP_{rev} = 295.15/20 = 14.76; the heater has COP = 1

η_{II} = 1/14.76 = 20/295.15 = **6.78 %**

Exam tip: First-law efficiency of a resistance heater is 100 %, yet it wastes over 93 % of the exergy of electricity.

Q11 **Answer: (B) zero**

Concept: General relation: $c_p - c_v = T\, v\, \beta^2/\kappa$ (β = volume expansivity, κ = isothermal compressibility).

For an incompressible substance v is constant, so $\beta = 0$ and $c_p - c_v = 0$.

So liquids and solids have a single specific heat c.

Exam tip: $c_p \geq c_v$ always, because $\kappa > 0$ for every stable substance.

Q12 **Answer: (B) 120.1 kJ**

Concept: Gouy–Stodola theorem: $I = T_0 S_{gen}$, with T_0 in kelvin.

I = 300.15 × 0.4

I = **120.1 kJ** (≈ 120 kJ)

Exam tip: Using 27 instead of 300.15 gives 10.8 kJ, the trap option.

Q13 **Answer: 732**

Concept: Same heat flux through both layers; find q from the total resistance, then step through the first layer.

$R'' = 0.2/1.2 + 0.1/0.15 = 0.1667 + 0.6667 = 0.8333\ m^2K/W$; $q = 840/0.8333 = 1008\ W/m^2$

$T_i = 900 - 1008 \times 0.1667 =$ **732 °C**

Exam tip: Check from the other side: 60 + 1008 × 0.6667 = 732 °C ✓.

Q14 **Answer: (B) always decrease the heat loss**

Concept: Heat loss versus outer radius peaks at $r_c = k/h$.

If the bare radius already exceeds r_c, we are on the falling side of the curve: every extra layer reduces heat loss.

This is the normal case for steam pipes (r_c of a few mm to a few cm).

Exam tip: The 'increase first' behaviour matters only for thin wires and small tubes.

Q15 **Answer: 305.5 (accept 302.4 to 308.6)**

Concept: Spherical shell: $Q = 4\pi k\, r_1 r_2 (T_2 - T_1)/(r_2 - r_1)$.

r_1 = 0.5 m, r_2 = 0.55 m, $\Delta T = 25 - (-196) = 221$ K

$Q = 4\pi \times 0.02 \times 0.5 \times 0.55 \times 221/0.05 =$ **305.5 W**

Exam tip: At this heat gain about 305.5/199 000 ≈ 1.5 g of N_2 boils off per second ($h_{fg} \approx 199$ kJ/kg).

What next?

Day 42 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

Dig Career Thrust app

Telegram: daily PDFs

Next, Day 43/90: Heat Transfer, Fins & Unsteady Conduction. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.